

September 2025 Exam

EC400

Probability and Statistical Inference

Suitable for all candidates

Instructions to candidates

This paper contains SEVEN questions. Answer ALL questions.

At the top of your answer scripts, please write the following:

- Your full name;
- Your class group; and
- Your class teacher's name.

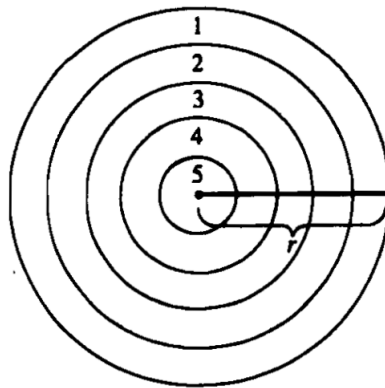
Time Allowed: *2 hours*

You are supplied with: *Standard Normal and Student's t tables*

Calculators: *Calculators are NOT allowed in this examination*

1. **Total: 15 points**

- (a) Let $(\Omega, \mathcal{F}, P)$ represent a probability space with probability measure P . Let the notation “ $\emptyset$ ” denote the null set, and “ $A \setminus B$ ” the set of elements in A but not B for any events $A, B \in \mathcal{F}$.
- State the axioms of probability, and prove that $P(\emptyset) = 0$. **(4pts)**
 - Prove that $P(A \setminus B) = P(A) - P(A \cap B)$. **(3pts)**
- (b) A game of darts is played by throwing a dart at a board and receiving a score corresponding to the number assigned to the region in which the dart lands. Referring to the figure below, we see that the innermost circle is worth 5 points and the outermost ring 1 point. Further, we see that the dart board has radius r and the distance between rings is $r/5$. For a novice player, it seems reasonable to assume that the probability of the dart hitting a particular region is proportional to the area of the region. Thus, a bigger region has a higher probability of being hit. We assume the dart board is always hit on any given throw.



- What is the probability that a novice player will score exactly 1 point on a given throw? **(3pts)**
- Let random variable X denote the number of points scored in a given throw of a dart by a novice player. Explain with detailed reasoning why the probability mass function (PMF) of X is given by:

$$P(X = i) = \frac{(6 - i)^2 - (5 - i)^2}{5^2},$$

for $i = 1, \dots, 5$, and 0 otherwise. Argue that $P(X = i)$ is a valid PMF independently of A , where $A = \pi r^2$, is the size of the dart board. **(5pts)**

2. Total: 15 points

Long Shorts, Inc., has two trading desks, A and B . Desk A executes $2/3$ of all trades, and desk B executes $1/3$. The probability that a trade results in a loss differs by desk: 5% of trades from A result in a loss, while 10% of trades from B result in a loss.

- (a) Let L denote the event that a randomly chosen trade results in a loss. Compute $P(L)$ using the law of total probability. **(3pts)**
 - (b) A trade is chosen at random and is found to have resulted in a loss. Compute the probability that it was executed by desk B , i.e. find $P(B | L)$. **(4pts)**
 - (c) Consider the two-desk system in general terms: $P(A) = p$, $P(B) = 1 - p$, $P(L | A) = \alpha$, $P(L | B) = \beta$, with $0 < p, \alpha, \beta < 1$. Show algebraically that if $\beta > \alpha$, then $P(B | L) > P(B)$, and explain why knowing that a trade resulted in a loss raises our belief that it was executed by desk B (relative to the unconditional probability that it was executed by desk B). **(5pts)**
 - (d) Suppose *Long Shorts, Inc.* introduces a risk audit that only examines trades that resulted in losses. Let R denote the event that a trade is selected for audit. Do you agree or disagree with the claim that $P(B | R) = P(B | L)$? Explain. **(3pts)**
-

3. Total: 15 points

Let $X \sim \text{Exponential}(\lambda)$ with rate parameter $\lambda > 0$. In other words, the probability density function (PDF) of X is given by:

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0 \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Derive the moment-generating function (MGF) of X , say $M_X(t)$, and state the values of t for which $M_X(t)$ is well-defined. **(3pts)**
- (b) Using the MGF, obtain the expected value and variance of X . **(3pts)**
- (c) Derive the cumulative distribution function (CDF) of X . **(3pts)**
- (d) Let $Z \sim \text{Bernoulli}(p)$. Find the expected value and variance of Z . **(3pts)**
- (e) Now let $Y = \mathbb{1}(X \leq c)$ where c is a positive constant, and $\mathbb{1}(\cdot)$ is the indicator function. Find the expected value and variance of Y . **(3pts)**

4. Total: 15 points

Suppose the random variable X has probability density function (PDF) given by:

$$f_X(x) = \begin{cases} 2x, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Let $Y = \sqrt{X}$ and define θ to be the fourth (raw) moment of Y .

(a) Find the PDF of Y using the change-of-variable formula. **(4pts)**

(b) Suppose we have a single observation: X . Consider the estimator

$$\hat{\theta} = Y^4.$$

Show that the mean squared error (MSE) of $\hat{\theta}$ is $1/12$. **(4pts)**

(c) Now suppose we have a random sample of size n : $X_1, \dots, X_n$. Consider the estimator

$$\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n (Y_i^4 + k),$$

where $k \in \mathbb{R}$ is a fixed constant. Compute the MSE of $\hat{\theta}_n$ in terms of its bias, the sample size n , and $\text{Var}(\hat{\theta})$. **(4pts)**

(d) Discuss the behaviour of the MSE of $\hat{\theta}_n$ as $n \rightarrow \infty$. Under what conditions on k is $\hat{\theta}_n$ consistent? **(3pts)**

5. Total: 10 points

Suppose $X_i \stackrel{iid}{\sim} \text{Geometric}(\pi)$ for $i = 1, \dots, n$, where π is unknown. In other words, the probability mass function of each of the X_i variables is given by

$$\Pr(X_i = x) = \begin{cases} (1 - \pi)^{x-1} \pi & x = 1, 2, \dots \\ 0 & \text{otherwise.} \end{cases}$$

Derive the maximum likelihood estimator for π given a sample $\mathbf{x} = (x_1, \dots, x_n)'$.

(You do **not** need to verify the second-order condition.) **(10pts)**

6. **Total: 20 points**

- (a) i. Congratulations! You have been hired to teach a class on hypothesis testing for BSc Economics students at LSE. You are preparing your class notes, in which you need to list the key steps involved in carrying out a hypothesis test. What do you write? (Provide 1-2 lines of explanation for each step.)

(5pts)

- ii. Your students love your teaching notes! However, they completely forget your guidance when they do their end-of-term empirical research papers. For instance, here is an excerpt from a randomly chosen student's paper:

*We want to see if **BrainJuice** increases IQ. We measured 16 people's IQ before and after drinking **BrainJuice**. The mean IQ before was 110, after was 115. We assumed normality, ran a one-sided test, and got $p = 0.02$. In other words, there is only a 2% chance that the null is true. Therefore, we conclude that **BrainJuice** increases IQ.*

Sigh! Identify at least 5 distinct mistakes and/or ambiguities in this student's work. Provide the student with constructive feedback in each case.

(5pts)

- iii. As you flip through the various research papers, you find a much better submission, from which the following extract immediately catches your attention:

Suppose $H_0 : \theta = 0$ and $H_1 : \theta \neq 0$. Power is defined as:

$$P(\text{Reject } H_0) = P(\text{Reject } H_0 | H_0 \text{ is true})P(H_0 \text{ is true}) \\ + P(\text{Reject } H_0 | H_0 \text{ is false})P(H_0 \text{ is false})$$

Comment on the above extract. Is the student's algebra correct? If not, identify the mistake(s), write down the correct algebraic definition of the power of a test, and explain why the student's analysis might be misleading.

(5pts)

- (b) The random variable X is normally distributed with mean μ and variance $\sigma^2 = 4$. A sample of $n = 16$ independent realisations of X was collected and the sample mean was $\bar{x} = 20.5$. Find a 90% confidence interval for μ . (You may state your answers as simplified expressions rather than exact numerical values.) **(5pts)**

7. Total: 10 points

(a) Give a precise definition for: **(3pts)**

- i. a χ_n^2 random variable;
- ii. a t_ν random variable; and
- iii. an $F_{p,q}$ random variable.

(b) Suppose $Z \sim \text{Normal}(0, 1)$ and define $U = Z^2$. Consider the ratio

$$T = \frac{Z}{\sqrt{U/1}}.$$

Does T follow the t distribution? Explain your reasoning. **(3pts)**

(c) Suppose $U \sim \chi_k^2$ and $V \sim \chi_m^2$ are independent. Consider the ratio

$$W = \frac{U/m}{V/k}.$$

Does W follow the F distribution? If so, specify the degrees of freedom; if not, explain how its stochastic behaviour is related to the F -distribution. **(4pts)**

END OF EXAM PAPER

TABLE FD.1 Observations on Income and Education, 20 Observations

I = Observation,
 Y = Income.

Source: Data are artificial.

APPENDIX G



STATISTICAL TABLES

TABLE G.1 Cumulative Normal Distribution. Table Entry Is $\Phi(z) = \text{Prob}[Z < z]$

[illegible]

TABLE G.2 Percentiles of the Student's t Distribution. Table Entry Is x Such that $\text{Prob}[t_n \leq x] = P$

n	.750	.900	.950	.975	.990	.995
1	1.000	3.078	6.314	12.706	31.821	63.657
2	.816	1.886	2.920	4.303	6.965	9.925
3	.765	1.638	2.353	3.182	4.541	5.841
4	.741	1.533	2.132	2.776	3.747	4.604
5	.727	1.476	2.015	2.571	3.365	4.032
6	.718	1.440	1.943	2.447	3.143	3.707
7	.711	1.415	1.895	2.365	2.998	3.499
8	.706	1.397	1.860	2.306	2.896	3.355
9	.703	1.383	1.833	2.262	2.821	3.250
10	.700	1.372	1.812	2.228	2.764	3.169
11	.697	1.363	1.796	2.201	2.718	3.106
12	.695	1.356	1.782	2.179	2.681	3.055
13	.694	1.350	1.771	2.160	2.650	3.012
14	.692	1.345	1.761	2.145	2.624	2.977
15	.691	1.341	1.753	2.131	2.602	2.947
16	.690	1.337	1.746	2.120	2.583	2.921
17	.689	1.333	1.740	2.110	2.567	2.898
18	.688	1.330	1.734	2.101	2.552	2.878
19	.688	1.328	1.729	2.093	2.539	2.861
20	.687	1.325	1.725	2.086	2.528	2.845
21	.686	1.323	1.721	2.080	2.518	2.831
22	.686	1.321	1.717	2.074	2.508	2.819
23	.685	1.319	1.714	2.069	2.500	2.807
24	.685	1.318	1.711	2.064	2.492	2.797
25	.684	1.316	1.708	2.060	2.485	2.787
26	.684	1.315	1.706	2.056	2.479	2.779
27	.684	1.314	1.703	2.052	2.473	2.771
28	.683	1.313	1.701	2.048	2.467	2.763
29	.683	1.311	1.699	2.045	2.462	2.756
30	.683	1.310	1.697	2.042	2.457	2.750
35	.682	1.306	1.690	2.030	2.438	2.724
40	.681	1.303	1.684	2.021	2.423	2.704
45	.680	1.301	1.679	2.014	2.412	2.690
50	.679	1.299	1.676	2.009	2.403	2.678
60	.679	1.296	1.671	2.000	2.390	2.660
70	.678	1.294	1.667	1.994	2.381	2.648
80	.678	1.292	1.664	1.990	2.374	2.639
90	.677	1.291	1.662	1.987	2.368	2.632
100	.677	1.290	1.660	1.984	2.364	2.626
∞	.674	1.282	1.645	1.960	2.326	2.576